

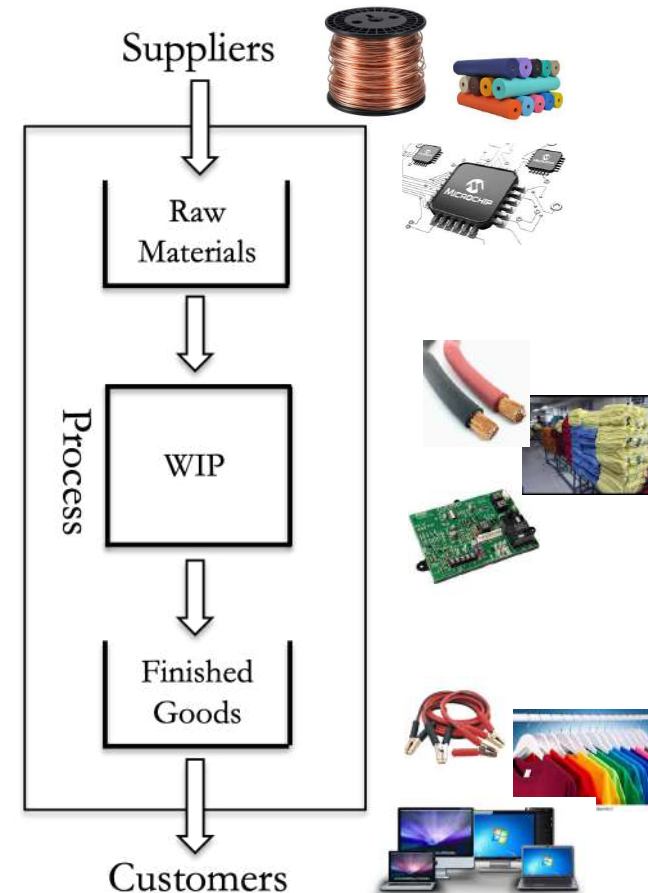
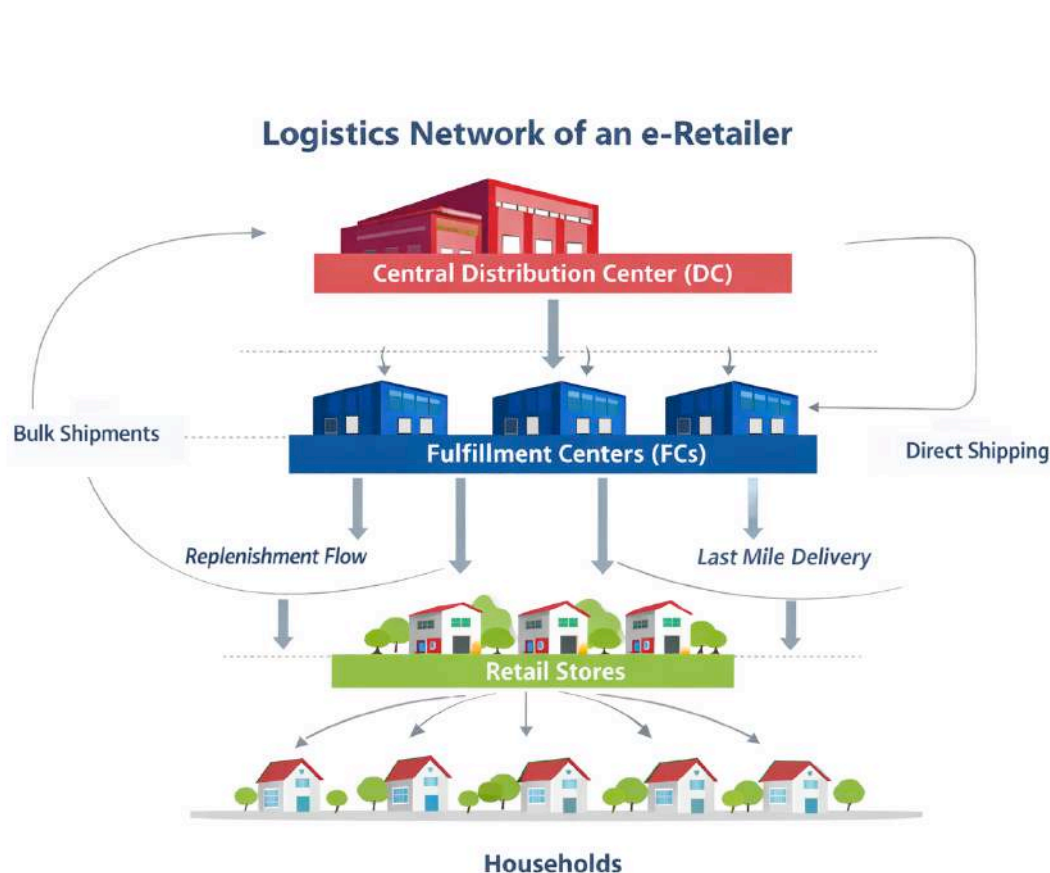
Operations Management

Session 6: Foundations of Inventory Management

Outline:

- Fundamentals
 - Why and where is inventory held?
 - Why is inventory management important?
 - Commonly used performance metrics
 - Inventory costs calculations
 - Basics of inventory management
 - Cycle Counting, Inventory Accuracy
 - ABC Analysis
- Inventory Management Strategies
 - Economic order quantity (EOQ)
 - Newsvendor problem
 - (Q,R) policies
- LL Bean

Inventory is the stock or store of an item or a resource used by an organization



Why Do Companies Hold Inventory?

➤ **Absorb market variability**

- ✓ To meet anticipated customer demand
 - ✧ Pipeline Stock – e.g. inventory shipped, inventory processed or waiting in a factory.
 - ✧ Seasonal inventory
- ✓ To protect against stock-outs
 - ✓ Safety stocks
- ✓ To allow for smooth and flexible production operations
 - ✧ WIP and Buffers
- ✓ To hedge against inflation, price increases, tariffs
 - ✧ Speculative inventory

➤ **Economies of scale**

- ✓ To take advantage of quantity discounts
 - ✧ Full truckload inventory
- ✓ To take advantage of economic order cycles
 - ✧ Cycle inventory

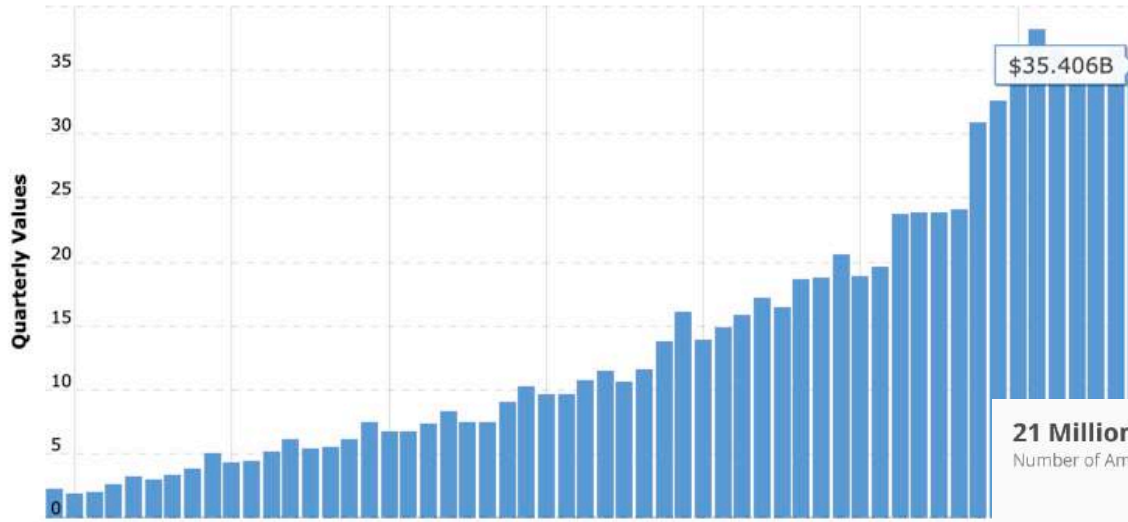
Importance of Inventory Management

- Inventories represent a major commitment of monetary resources
- Example: 40% of Best Buy's assets comprise of inventory
- Inventories affect virtually all aspects of a company's daily operations
- Inventories are a major competitive weapon for many companies
- Example: Dell, Wal-Mart, Zara.

Importance of Inventory Management



From: 12/31/2009 To: 10/01/2023



Amazon Total Products*:

12,231,203

Amazon + Marketplace Total Products*:

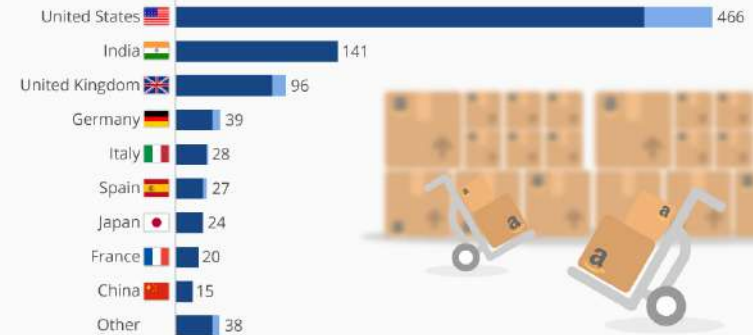
353,710,754

*Excludes Books, Media, Wine, and Services. As well, this analysis also excludes product variants.

21 Million M² Of Logistics Space Worldwide

Number of Amazon distribution locations worldwide in July 2019

■ Current ■ Future



© StatistaCharts

Source: MWPVL International

statista

Importance of Inventory Management

Comparison with stock market reaction to other corporate events

Operational events

Increase in capital expenditure	1.0%
Increase in R&D expenditure	1.4%
Effective TQM implementation	0.7%
Internal corporate restructuring	1.0%
Decrease in capital expenditure	-1.8%
Plant closing	-0.7%
Undersupply (shortfalls)	-7.2%

Information technology events

IT Investments	1.0%
B2C e-commerce	10.5%
B2B e-commerce	3.3%
IT problems	-1.7%

Marketing events

Change in firm name	0.7%
Brand leveraging	0.3%
Celebrity endorsement	0.2%
New product introduction	0.7%
Affirmative action awards	1.6%
Delay introduction of new products	-5.3%

Financial events

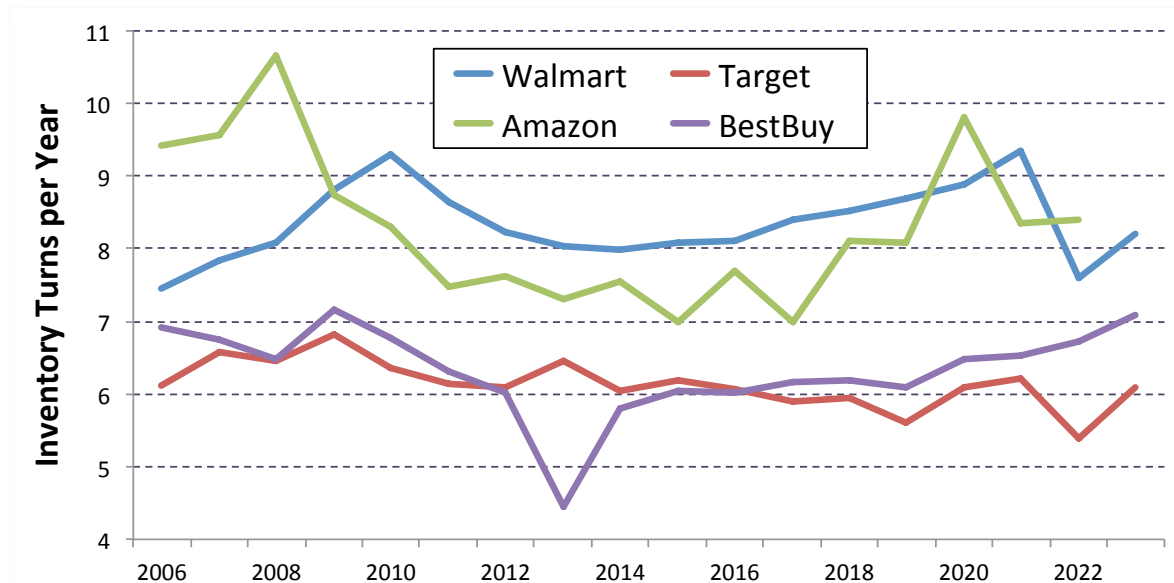
Stock splits	3.3%
Open market share repurchase	3.5%
Proxy contest	4.2%
Increasing financial leverage	7.6%
Decreasing financial leverage	-5.4%
Seasoned equity offerings	-3.0%

Inventory Measures

Average aggregate inventory value:

Average of the total value of all items held in inventory

$$\text{Inventory turns} = \frac{\text{Cost of Goods Sold per Year}}{\text{Average Aggregate Inventory Value}}$$



Example: Inventory Turns

➤ Example (Target: 2025):

- Inventory value = 12.313 B \$
- Cost of Goods Sold = 76.502 B \$/year
- Average time to turn a dollar (cost) to a dollar
= $12.313 / 76.502 \text{ year} = 0.161 \text{ years} = 58.7 \text{ days}$. (INV= λ CT Little's law)
- What is the financial significance of this time?
- Target improved this figure from 62 days in 2015.
- Target Inventory Turns in 2025 = $(1 / 0.161) = 6.22 \text{ Turns per year}$

➤ Example (Walmart: 2025):

- Inventory value = 56.44 B\$, COGS =
- Time to turn a dollar =
- Inventory Turns =

Inventory Costs

- **Ordering cost:** (per order/transaction)
 - cost incurred each time an order is placed with a supplier or production.
 - ✓ Shipping fees and transportation costs, inspection fees
- **Holding cost:** (per unit of inventory per unit time)
 - cost associated with maintaining an item in inventory until used/sold.
 - ✓ Interest on capital invested and insurance: 11% (6-20%),
 - ✓ Warehousing: 6% (3-10%),
 - ✓ Extra staffing: 4% (3-5%),
 - ✓ Material handling: 3% (1-4%),
 - ✓ Pilferage, damage and obsolescence: 3% (2-5%)
- **Stockout or shortage cost:** (per unit of lost sale)
 - occurs when the demand for an item exceeds its supply.
- **Item cost:** (per unit of inventory)
 - becomes relevant if a quantity discount is available.

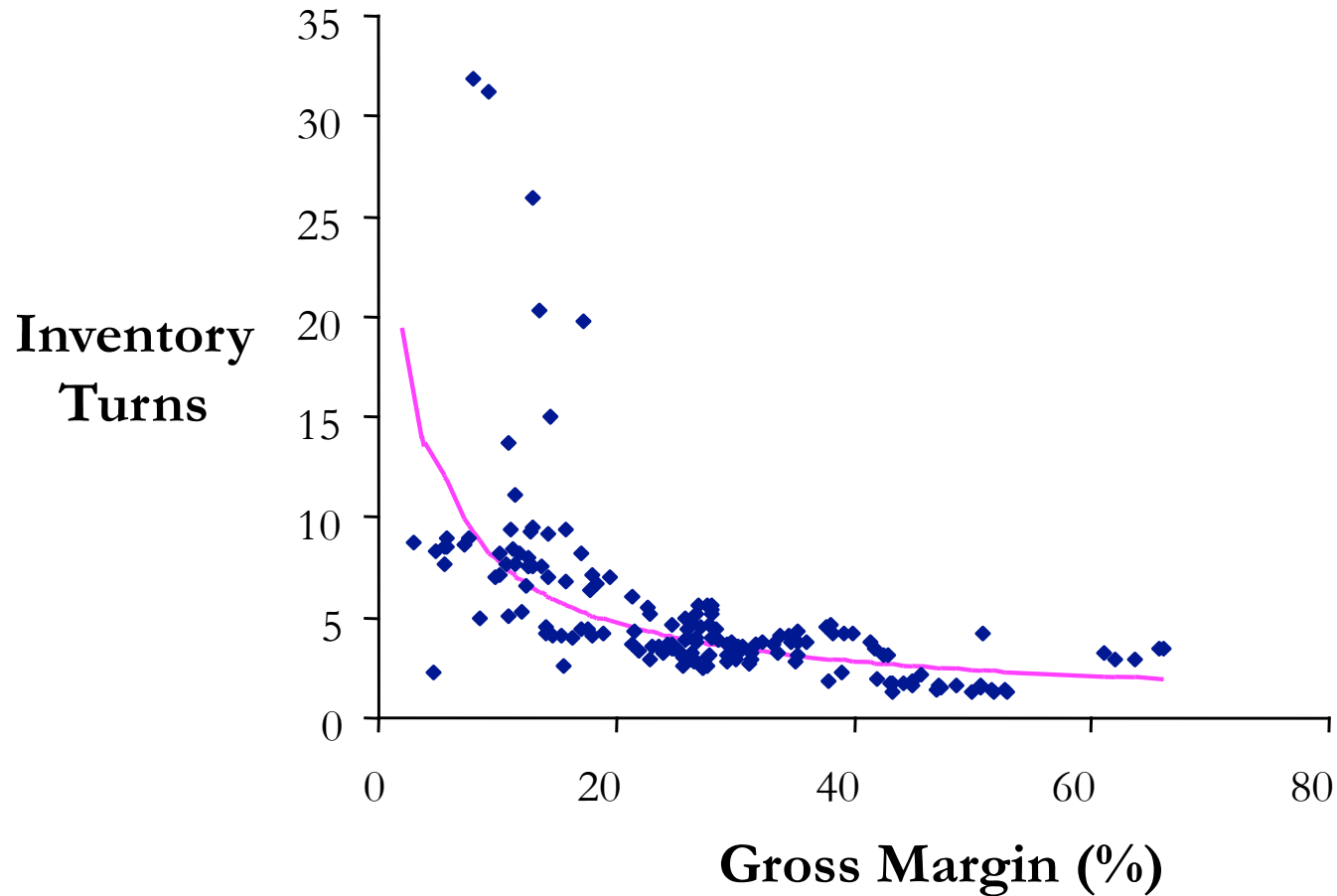
Inventory Turns: Financial Implications

- Consider a retailer whose inventory holding cost is 0.3 \$/\$/year
 - ✓ That is, to hold one unit of a good that costs 1 \$ for 1 year, we incur 30¢.
 - ✓ Say, this retailer turns inventory 4 times per year.
 - ✓ So, on average, each unit stays in inventory for $(1/4)$ of a year.
 - ✓ So, the cost that the retailer incurs just due to holding inventory is $30/4 = 7.5$ ¢ per \$.
 - ✓ In other words, when we buy a product for \$100 at this retailer, \$7.5 are paid towards the retailer's inventory holding costs.

- Target vs. Walmart (2025):
 - ✓ Target turns 6.22 per year for an inventory cost of $30/6.22 = 4.8$ ¢ per \$.
 - ✓ Walmart turns 9.1 per year for an inventory cost of $30/9.1 = 3.3$ ¢ per \$.

Inventory Turns

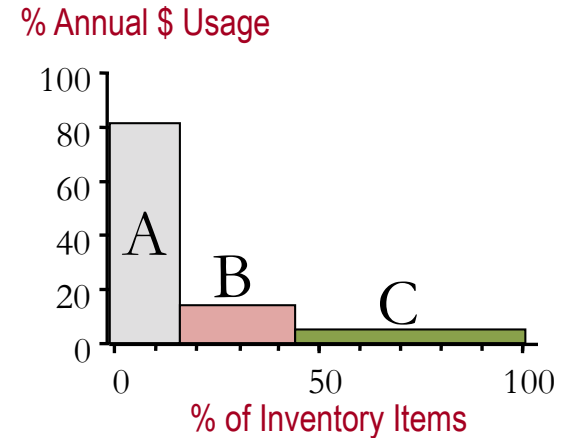
Inventory Turns Electronics and Computer Retailers



Managing Inventory

ABC Analysis

- Divides on-hand inventory into 3 classes
 - ✓ A class, B class, C class
 - ✓ Basis is usually annual \$ volume
 - ✧ \$volume = Annual demand x Unit cost
 - ✓ Policies based on ABC analysis
 - ✧ Develop class A suppliers more
 - ✧ Give tighter physical control of A items
 - ✧ Forecast A items more carefully



Class	% \$ Vol	% Items
A	80	15
B	15	30
C	5	55

Takeaway:

ABC is not about counting items, it is about allocating managerial attention.

- ✧ You should optimize A, manage B, and simplify C.
- ✧ Applying “sophisticated” policies uniformly is wasteful.
- ✧ Most inventory cost reductions come from doing a few things very well, not many things marginally better.

Inventory Accuracy

- Retail Industry: (Source: “Execution: The Missing Link in Retail Operations,” by Raman, DeHoratius, Ton)
- Leading Retailer:
 - ✓ Store Level SKUs (stock keeping units): 65% of records were inaccurate
 - ✓ Average error was 35 %
 - ✓ Despite scanning all sales and extensive IT (paradox)
- Another leading retailer:
 - ✓ 16% of items were reported “out of stock” by customers
 - they were actually in stock, wrong aisle, backroom
- Grocery stores: “regular tomatoes” and “organic tomatoes”
- Estimates of profit losses due to Inventory Inaccuracy and Misplaced SKUs – 10-25 % of current profits
- Automated forecasting system.

Cycle Counting

- Physically counting a sample of total inventory on a regular basis
 - ✓ Count items
 - ✓ Compare with records
 - ✓ Document inaccuracies and trace causes
- Used often with ABC classification
 - ✓ A items counted most often (e.g., daily)
- Another practice: shutting down the facility annually for inventory audits.

Inventory Models

➤ Economic Order Quantity (EOQ)

- ✓ Long lifecycle products with stable demand
- ✓ Low demand uncertainty
- ✓ Typically low unit margin
- ✓ Examples: Staples, never-out,
- ✓ Goal: Minimize costs subject to service level constraints

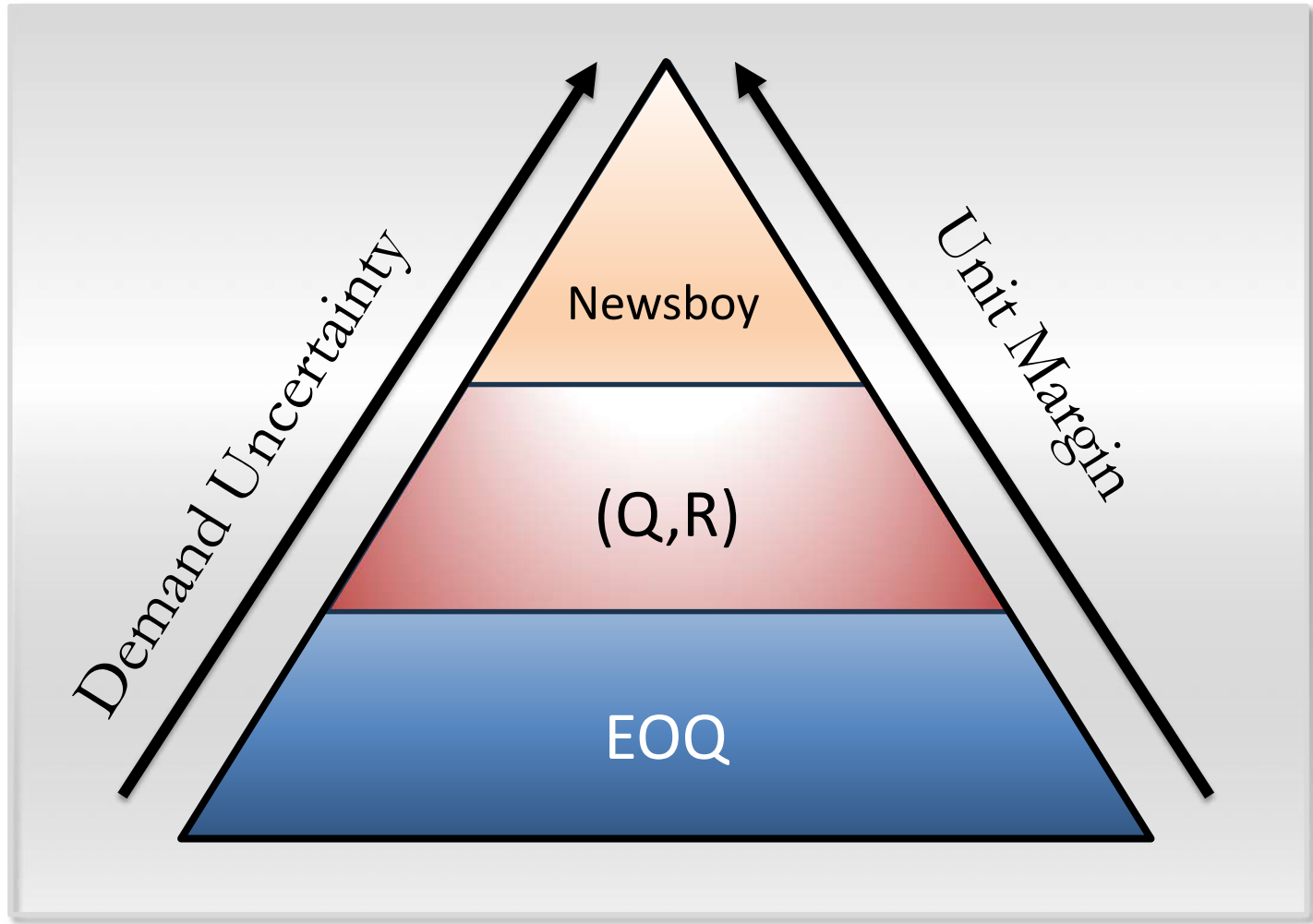


➤ Newsboy model

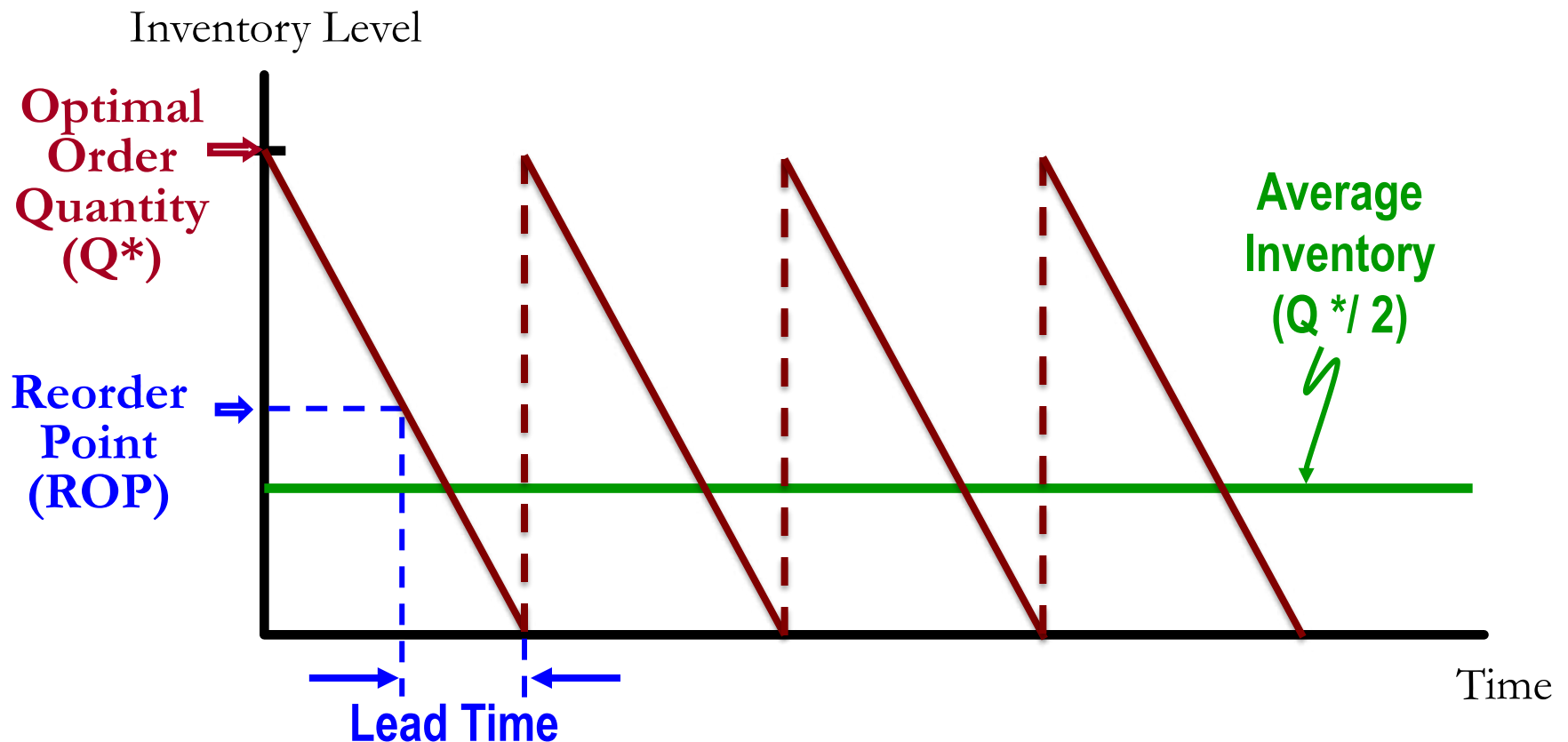
- ✓ Short lifecycle products (typically no re-stocking)
- ✓ Considerable demand uncertainty
- ✓ Typically higher unit margin
- ✓ Examples: fashion clothing, toys, computer games, ...
- ✓ Goal: Maximize (expected) profits
 - Managing tradeoff: overstocking & stockouts



Inventory Models



Economic Order Quantity (EOQ)



Economic Order Quantity (EOQ)

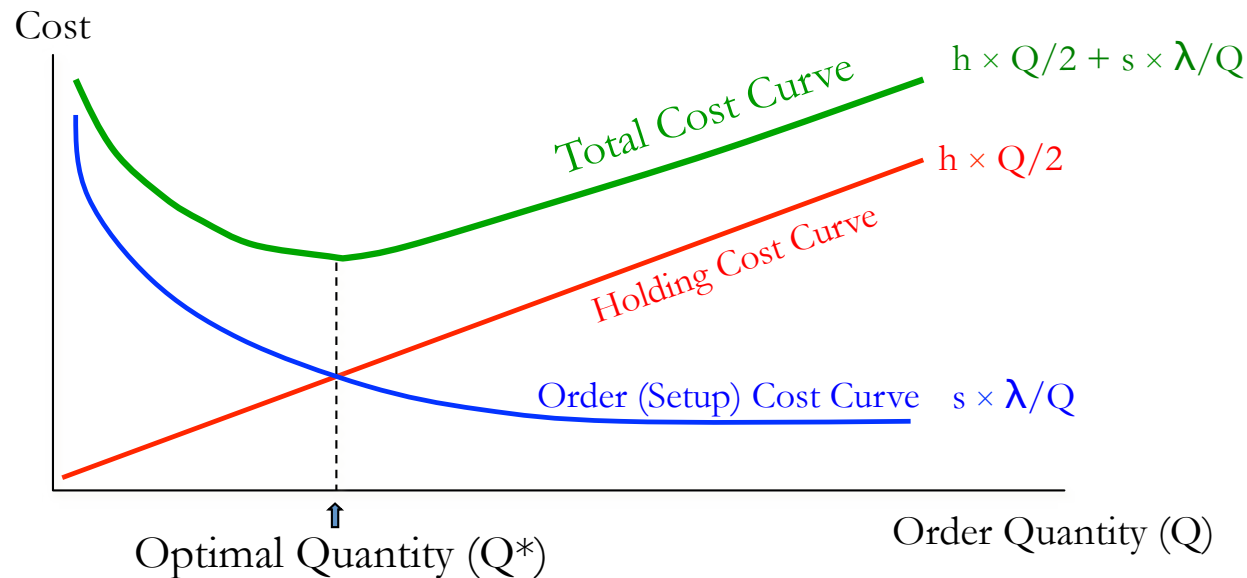
Model Assumptions

- ✓ Single product or item
- ✓ Demand rate known and constant
- ✓ Item produced in lots, or purchased in orders
- ✓ Each lot or order received in single delivery
- ✓ Lead time known and constant
- ✓ Ordering, or setup costs are constant
- ✓ No backorders are allowed
- ✓ No quantity discounts are allowed

Data, Cost Computation & EOQ

Data:

- ✧ λ = Demand (units) per period
- ✧ h = Holding cost per unit per period
- ✧ s = Ordering cost per order placed
- ✧ Q = Order quantity (units).



$$\text{Solution (EOQ): } Q^* = \sqrt{\frac{2 \lambda s}{h}}$$

Example: ABC Company

ABC sells industrial components for household appliances. A particular spindle has a weekly demand of 2,000 units. The lead time from the manufacturer is one week, and there is a fixed delivery cost of \$300 per shipment. Each spindle has a variable unit cost of \$10. The company's annual cost of capital is 20%. In addition, the accounting department estimates an additional holding cost of \$3 per unit per year for warehouse space and handling.

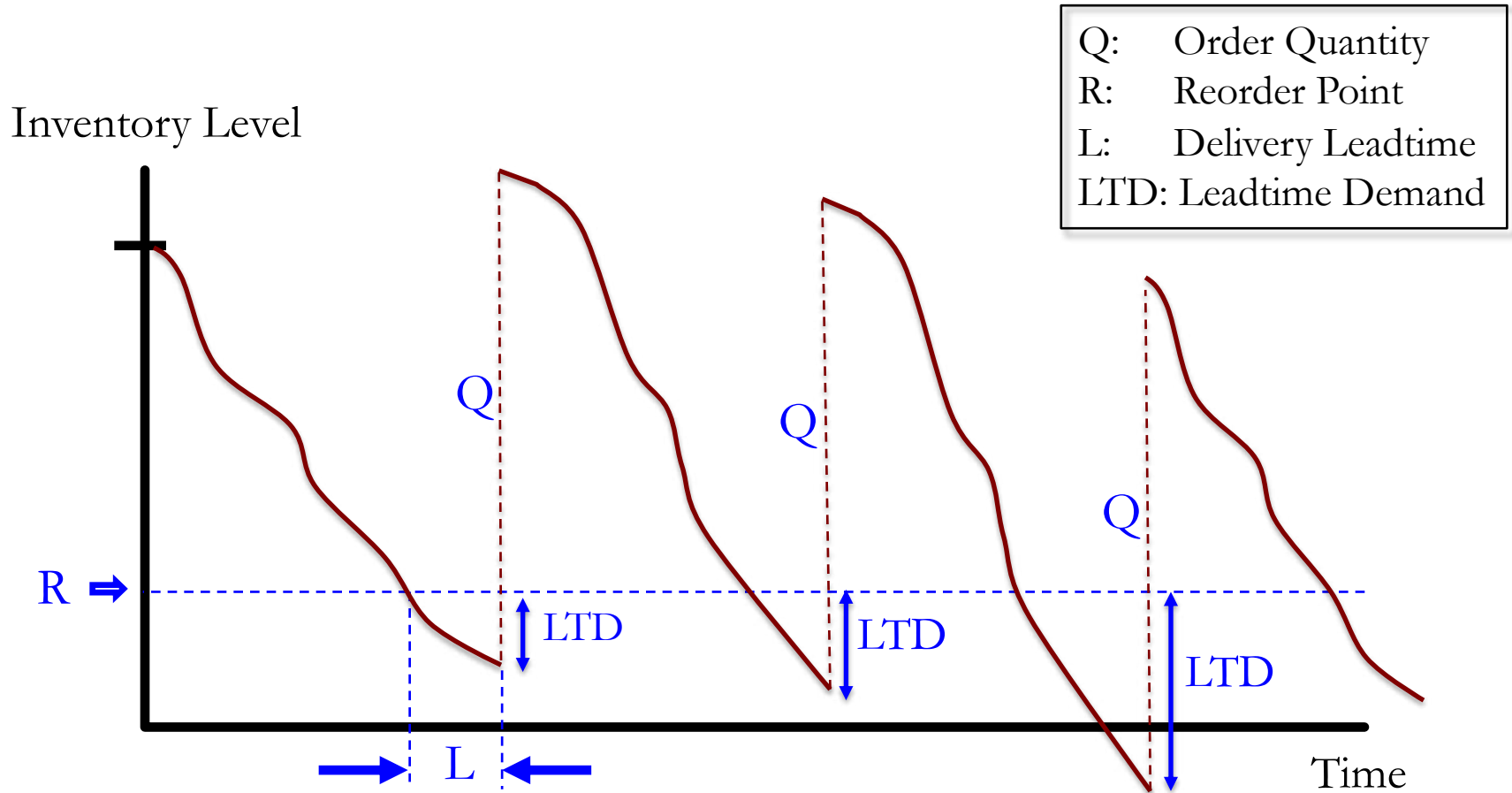
- a) Determine the economic order quantity and the corresponding annual cost.
- b) What is the average time (in days) that a spindle stays in the warehouse as inventory?
- c) What is the reorder point?
- d) What is the incremental annual cost if the company replenishes inventory monthly?

Example: Nurse Onboarding Training Program

A large hospital system hires registered nurses continuously, but new hires cannot be placed on the floor until they complete a mandatory 2--3 day onboarding program (EHR training, medication safety, compliance, badges). Because each onboarding cohort requires booking classrooms, scheduling educators, and setting up IT accounts, the hospital runs onboarding in batches. The hospital hires nurses at a steady rate of 520 nurses per year (about 10 per week). Each onboarding cohort entails a fixed cost of \$12,000 (educators, rooms, and IT provisioning). Nurses hired between cohorts are paid but cannot yet provide patient care; the company estimates the cost of an idle nurse to be \$55,000 per year in salary and benefits while waiting to be onboarded.

How should the hospital plan onboarding, i.e., how often should it hold onboarding sessions and how many nurses should it include in each cohort? What is the program's total annual cost (including both cohort setup costs and the cost of nurses waiting to be onboarded)?

The (Q,R) Policy (EOQ + Safety Stock for Random Demand)



Service Level & Safety Stock

How much & when to order

Demand can vary

- ✓ Lead Time Demand: Demand during leadtime
- ✓ Leadtime demand follows normal distribution (forecasting)
- ✓ Other EOQ assumptions apply

Consider service level & safety stock

- ✓ Service level = $1 - \text{Probability of stockout}$
- ✓ Higher service level means more safety stock
 - More safety stock means higher ROP
 - $\text{ROP} = \text{mean of leadtime demand} + \text{safety stock}$

Two Fundamental Tradeoffs

- ✧ **Q:** Inventory holding versus ordering costs
Economic Order Quantity (EOQ). ✓
- ✧ **R:** Inventory holding versus stockout costs
Optimal service level determined through Newsvendor solution.

Two possibilities:

- ✓ If $LTD > R$ then Stockout occurs.
- ✓ If $LTD \leq R$ then no Stockout occurs.

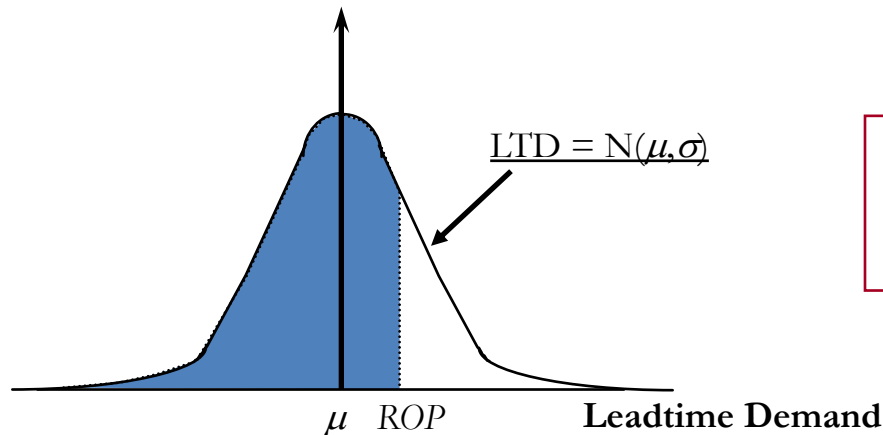
So choose smallest R so that

$\text{Prob}(LTD \leq R) \geq \text{“Optimal” Service Level.}$

This is R^* !

Safety Stock Computation

- Leadtime demand is a normal distribution with mean μ and standard deviation σ .
- Decide on the target performance (Service Level) SL
 - ✓ e.g. SL = 99% means that 99% of the time the leadtime demand is less than the ROP.
- Find the corresponding level of inventory ROP
 - ✓ Likelihood that leadtime demand is smaller than ROP is SL
 - ✓ $ROP = \mu + \text{Safety Stock}$



Using Excel the ROP is
 $ROP = \text{NORMINV}(\text{SL}, \mu, \sigma)$

Probabilistic Model: Examples

➤ Demand during leadtime for one brand of TV is normally distributed with a mean of 36 TVs and a standard deviation of 15 TVs. What safety stock should be carried for a 90% service level? What is the appropriate reorder point?

➤ Based on available information, the daily demand for smart phone averages 10 units (normally distributed), with a standard deviation of 1 phone. The leadtime is exactly 5 days. Management wants a 97% service level. What safety stock should be carried? What is the appropriate reorder point?



ANNUAL SALES: \$599M (1990), \$1.8B (2021)

EMPLOYEES: 5,500 year-round, 10,000 during winter holidays including 2,500 CSR's (2021)

CALL CENTERS: 2 in Maine, 14 million orders in 2016

STOREFRONTS: 56 retail stores & outlets + flagship store (US),
24 stores in Japan, 8 stores in Canada

DISTRIBUTION: 1,000,000 square-foot DC, capacity of 50,000 customer orders per day, can store 10 million units of merchandise, shipped 14.5 million packages in 2016.

PRODUCTS: 6,000 items (1990), 140,000 items (2016)

Source: www.llbean.com

CATALOGUE INDUSTRY (2016):

- 9.8B catalogues mailed. Avg. cost \$1 per catalogue.
- 42% read catalogues received in mail
- 100M catalogue purchase
- Direct mail response: 5.3% house file and 2.9% prospect list.
- Article: Millennials are more interested in catalogs than your grandmother

What are L.L. Bean's inventory management challenges?

Struggling to hold the right amount and right kind of inventory:

- \$10M annually in left over stock
- \$11M annually in lost sales, backorders

Is demand forecast particularly challenging for L.L. Bean compared to other retailers?

Newsvendor Model

Every morning the newsvendor must decide the number of newspapers **Q** to buy.

The newsvendor pays a unit cost of **\$c** per newspaper and sells them at a price of **\$p** per unit.

However, daily demand **D** is random and unsold newspapers are discarded at a salvage value **\$s**.



Optimization Problem:

Find the value of Q that maximizes daily expected profits:

$$\text{Profits} = E[p * \text{Min}\{\mathbf{D}, Q\} - c * Q + s * \text{Max}\{0, Q - \mathbf{D}\}]$$

The Newsvendor Problem

Fundamental Tradeoff

Too much of the wrong stuff versus too little of the right stuff

Other examples:

- ✓ Retail stores
- ✓ Airline seats (revenue or yield management)
- ✓ Hotels
- ✓ Rental cars
- ✓ Call center staffing
- ✓ Plant sizing

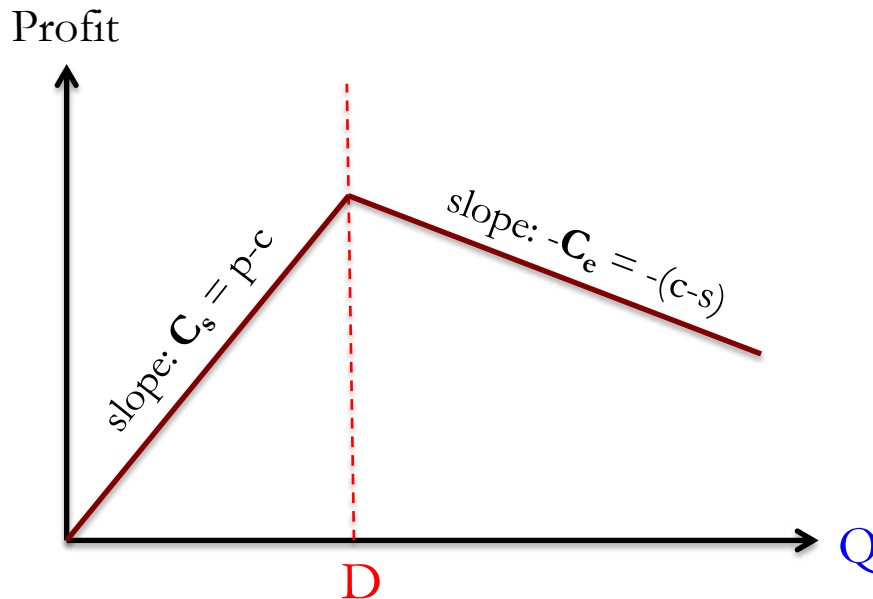
Notation and Profit Function

NOTATION:

- ✓ p = retail price
- ✓ c = wholesale cost
- ✓ s = salvage value
- ✓ D = demand (random variable)
- ✓ Q = order quantity (decision variable)

PROFIT FUNCTION:

$$\text{Profit}(Q) = p \times \underbrace{\text{Min}\{D, Q\}}_{\text{sales}} - c \times Q + s \times \underbrace{\text{Max}\{0, Q - D\}}_{\text{left-over inventory}}$$



C_e = cost of excess = $c - s$

- ✧ The cost of ordering one more unit than what you would have ordered had you known demand.
- ✧ Include any purge cost if additional expense is required to dispose of leftover inventory

C_s = cost of shortage = $p - c$

- ✧ The cost of ordering one fewer unit than what you would have ordered had you known demand.
- ✧ Include goodwill cost or penalty cost associated to unsatisfied demand.

Timeline and Economics



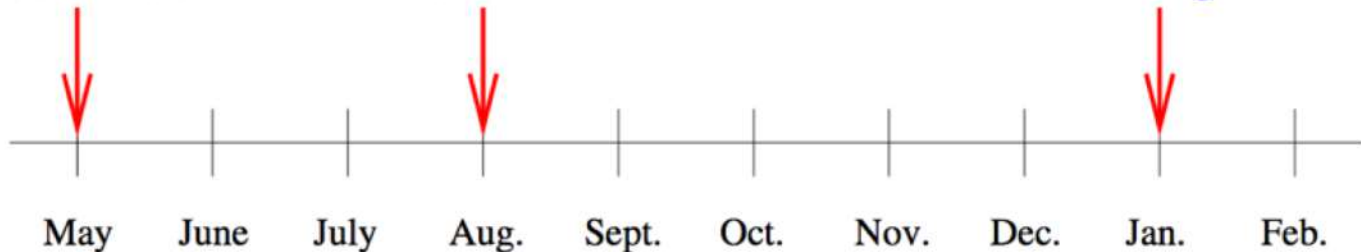
Economics:

- ✧ Price \$45
- ✧ Wholesale cost \$25
- ✧ Discounted sales price \$15
- ✧ Forecast demand 12,000 units

Generate forecast
and submit order

Receive
order

Left over units
salvaged



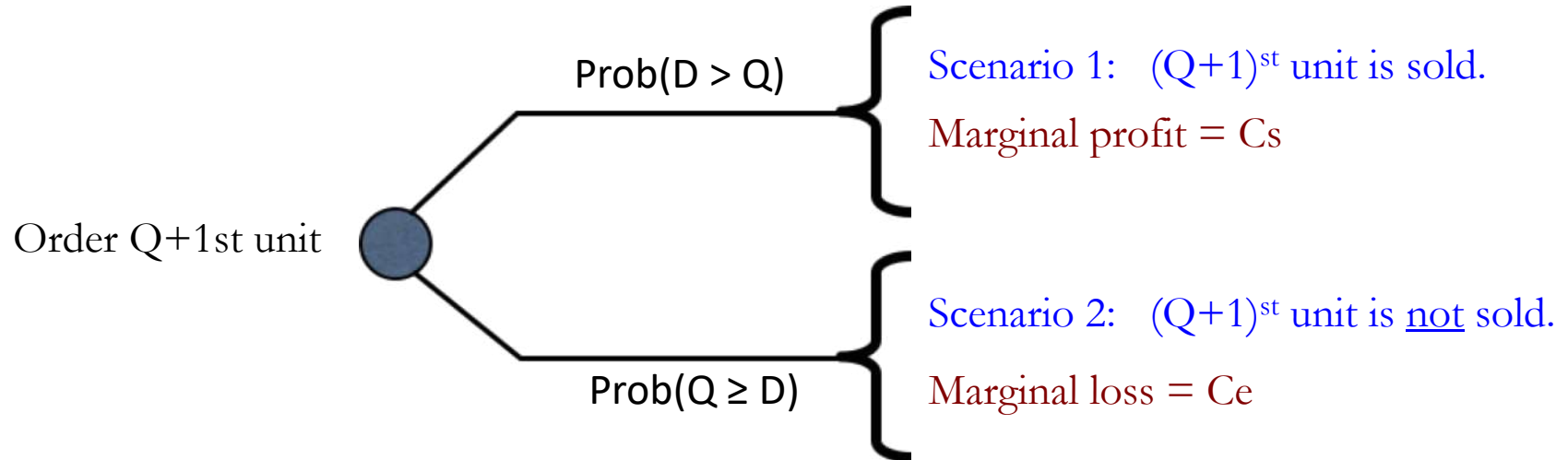
Fall Selling Season

No “Quick Response” procurement available, single order commitment (Newsvendor problem)

L.L. Bean's Expected Profit Function

Balancing marginal gain and marginal loss

Suppose current order quantity is Q . Consider $(Q + 1)^{\text{st}}$ unit.



✧ Expected marginal gain
 $= C_s \times \text{Prob}(D > Q) = C_s \times (1 - \text{Prob}(D \leq Q))$

✧ Expected marginal loss
 $= C_e \times \text{Prob}(D \leq Q)$

✧ Expected net marginal gain
 $= C_s \times (1 - \text{Prob}(D \leq Q)) - C_e \times \text{Prob}(D \leq Q)$

Recall:
 $\text{Prob}(D > Q) = 1 - \text{Prob}(D \leq Q)$

General Solution: Critical Fractile

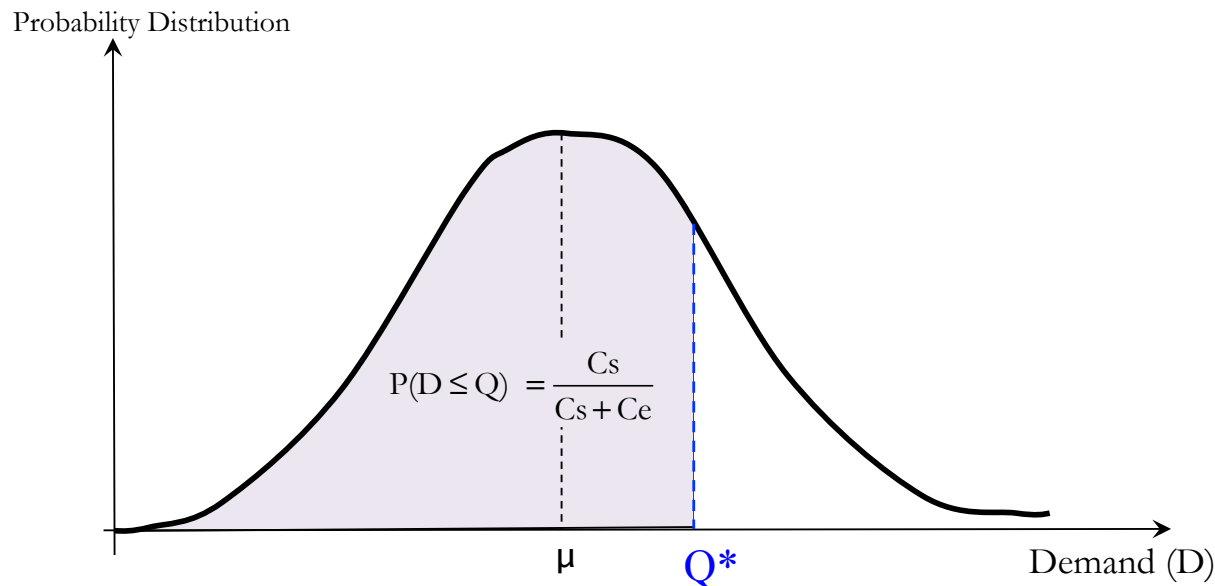
Optimality Principle:

Expected net marginal gain = 0

Expected marginal gain = Expected marginal loss

$$C_s \times (1 - \text{Prob}(D \leq Q)) = C_e \times \text{Prob}(D \leq Q)$$

$$\text{Prob}(D \leq Q) = \frac{C_s}{C_s + C_e} = \text{Critical Fractile (CF)} = \text{Optimal Service Level (SL*)}$$



L.L. Bean Solution: Normal Distribution

Computing Expected Profit

$$\begin{aligned}\text{Expected Profit (Q)} &= p \times E[\text{Min}\{\text{D}, \text{Q}\}] - c \times \text{Q} + s \times E[\overbrace{\text{Max}\{0, \text{Q} - \text{D}\}}^{\text{Q} - \text{Min}\{\text{D}, \text{Q}\}}] \\ &= (C_s + C_e) \times E[\text{Min}\{\text{D}, \text{Q}\}] - C_e \times \text{Q}\end{aligned}$$

Loss Function: $L(z)$ (see file Normal_Tables.pdf available on Canvas)

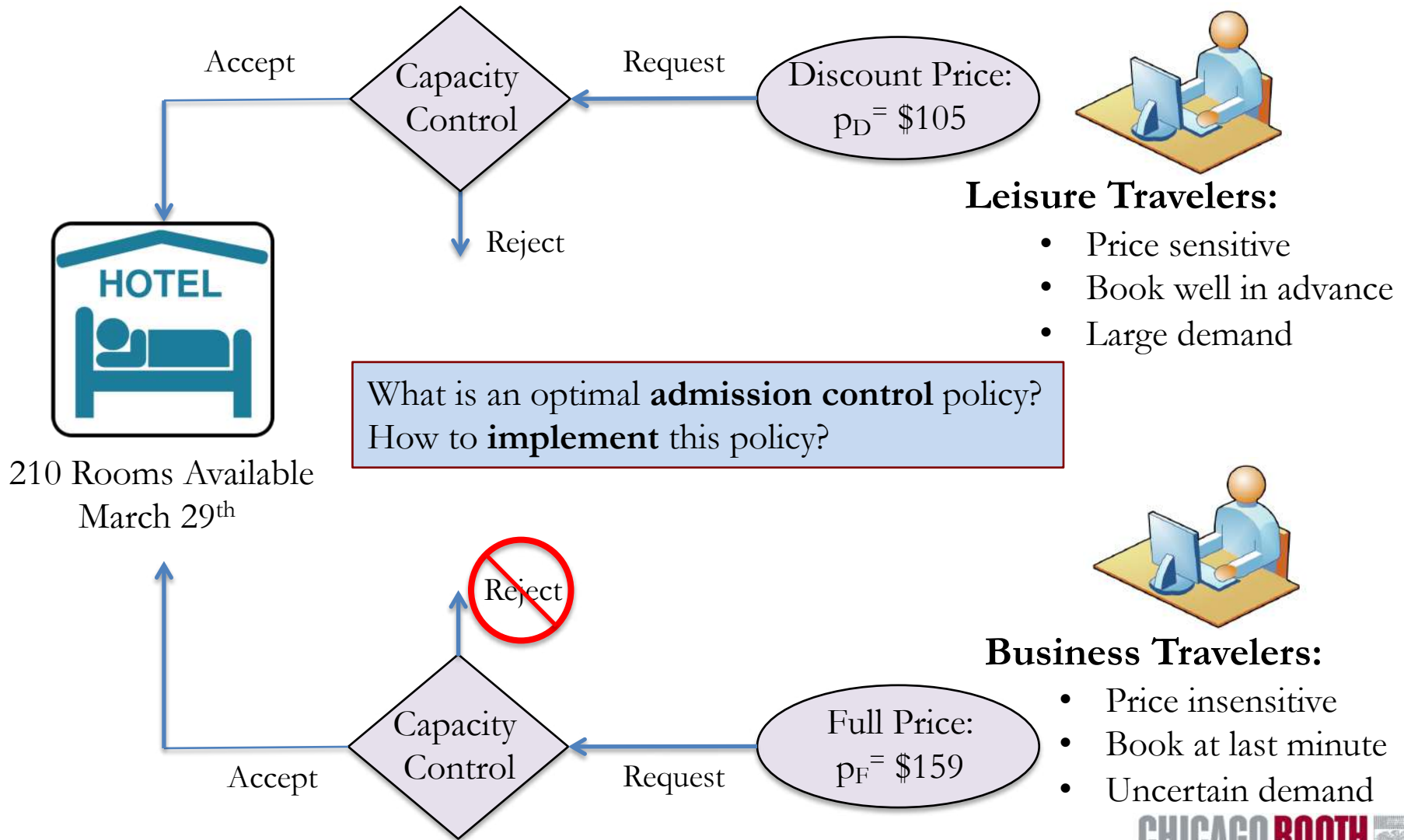
Let $D \sim \text{Normal}(\mu, \sigma)$ and set the Z-score at level Q to be $z = (Q - \mu)/\sigma$. Then,

- ✓ $E[\text{Min}\{D, Q\}] = \mu - \sigma \cdot L(z)$.
- ✓ In Excel: $L(z) = \text{Normdist}(z, 0, 1, 0) - z \times (1 - \text{Normsdist}(z))$.

Z-Chart & Loss Function

Z	F(Z)	L(Z)	Z	F(Z)	L(Z)	Z	F(Z)	L(Z)	Z	F(Z)	L(Z)
-3.00	0.0013	3.000	-1.48	0.0694	1.511	0.04	0.5160	0.379	1.56	0.9406	0.026
-2.96	0.0015	2.960	-1.44	0.0749	1.474	0.08	0.5319	0.360	1.60	0.9452	0.023
-2.92	0.0018	2.921	-1.40	0.0808	1.437	0.12	0.5478	0.342	1.64	0.9495	0.021
-2.88	0.0020	2.881	-1.36	0.0869	1.400	0.16	0.5636	0.324	1.68	0.9535	0.019
-2.84	0.0023	2.841	-1.32	0.0934	1.364	0.20	0.5793	0.307	1.72	0.9573	0.017
-2.80	0.0026	2.801	-1.28	0.1003	1.327	0.24	0.5948	0.290	1.76	0.9608	0.016
-2.76	0.0029	2.761	-1.24	0.1075	1.292	0.28	0.6103	0.274	1.80	0.9641	0.014
-2.72	0.0033	2.721	-1.20	0.1151	1.256	0.32	0.6255	0.259	1.84	0.9671	0.013
-2.68	0.0037	2.681	-1.16	0.1230	1.221	0.36	0.6406	0.245	1.88	0.9699	0.012
-2.64	0.0041	2.641	-1.12	0.1314	1.186	0.40	0.6554	0.230	1.92	0.9726	0.010
-2.60	0.0047	2.601	-1.08	0.1401	1.151	0.44	0.6700	0.217	1.96	0.9750	0.009
-2.56	0.0052	2.562	-1.04	0.1492	1.117	0.48	0.6844	0.204	2.00	0.9772	0.008
-2.52	0.0059	2.522	-1.00	0.1587	1.083	0.52	0.6985	0.192	2.04	0.9793	0.008
-2.48	0.0066	2.482	-0.96	0.1685	1.050	0.56	0.7123	0.180	2.08	0.9812	0.007
-2.44	0.0073	2.442	-0.92	0.1788	1.017	0.60	0.7257	0.169	2.12	0.9830	0.006
-2.40	0.0082	2.403	-0.88	0.1894	0.984	0.64	0.7389	0.158	2.16	0.9846	0.005
-2.36	0.0091	2.363	-0.84	0.2005	0.952	0.68	0.7517	0.148	2.20	0.9861	0.005
-2.32	0.0102	2.323	-0.80	0.2119	0.920	0.72	0.7642	0.138	2.24	0.9875	0.004
-2.28	0.0113	2.284	-0.76	0.2236	0.889	0.76	0.7764	0.129	2.28	0.9887	0.004
-2.24	0.0125	2.244	-0.72	0.2358	0.858	0.80	0.7881	0.120	2.32	0.9898	0.003
-2.20	0.0139	2.205	-0.68	0.2483	0.828	0.84	0.7995	0.112	2.36	0.9909	0.003
-2.16	0.0154	2.165	-0.64	0.2611	0.798	0.88	0.8106	0.104	2.40	0.9918	0.003
-2.12	0.0170	2.126	-0.60	0.2743	0.769	0.92	0.8212	0.097	2.44	0.9927	0.002
-2.08	0.0188	2.087	-0.56	0.2877	0.740	0.96	0.8315	0.090	2.48	0.9934	0.002
-2.04	0.0207	2.048	-0.52	0.3015	0.712	1.00	0.8413	0.083	2.52	0.9941	0.002
-2.00	0.0228	2.008	-0.48	0.3156	0.684	1.04	0.8508	0.077	2.56	0.9948	0.002
-1.96	0.0250	1.969	-0.44	0.3300	0.657	1.08	0.8599	0.071	2.60	0.9953	0.001
-1.92	0.0274	1.930	-0.40	0.3446	0.630	1.12	0.8686	0.066	2.64	0.9959	0.001
-1.88	0.0301	1.892	-0.36	0.3594	0.605	1.16	0.8770	0.061	2.68	0.9963	0.001
-1.84	0.0329	1.853	-0.32	0.3745	0.579	1.20	0.8849	0.056	2.72	0.9967	0.001
-1.80	0.0359	1.814	-0.28	0.3897	0.554	1.24	0.8925	0.052	2.76	0.9971	0.001
-1.76	0.0392	1.776	-0.24	0.4052	0.530	1.28	0.8997	0.047	2.80	0.9974	0.001
-1.72	0.0427	1.737	-0.20	0.4207	0.507	1.32	0.9066	0.044	2.84	0.9977	0.001
-1.68	0.0465	1.699	-0.16	0.4364	0.484	1.36	0.9131	0.040	2.88	0.9980	0.001
-1.64	0.0505	1.661	-0.12	0.4522	0.462	1.40	0.9192	0.037	2.92	0.9982	0.001
-1.60	0.0548	1.623	-0.08	0.4681	0.440	1.44	0.9251	0.034	2.96	0.9985	0.000
-1.56	0.0594	1.586	-0.04	0.4840	0.419	1.48	0.9306	0.031	3.00	0.9987	0.000
-1.52	0.0643	1.548	0.00	0.5000	0.399	1.52	0.9357	0.028			

Example: Hotel Reservation System



Hotel Reservation Example (cont'd)

Data:

- DISCOUNT PRICE: $\$p_D = \105
- FULL PRICE: $\$p_F = \159
- AVAILABLE CAPACITY: $C = 210$ units.
- FULL PRICE DEMAND: D_F . The following data is available:



210 Rooms Available
March 29th

Empirical Demand Distribution

Demand for rooms at full fare (Q)	# Days with Demand	Cumulative Probability $F(Q) = \text{Prob}\{D \leq Q\}$
0 - 70	12	0.098
71	3	0.122
72	3	0.146
73	2	0.163
74	0	0.163
75	4	0.195
76	4	0.228
77	5	0.268
78	2	0.285
79	7	0.341
80	4	0.374
81	10	0.455
82	13	0.561
83	12	0.659
84	4	0.691
85	9	0.764
86	10	0.846
above 86	19	1.000
Total	123	1.000

How many rooms to protect for business travelers?

LL Bean Forecasting Method

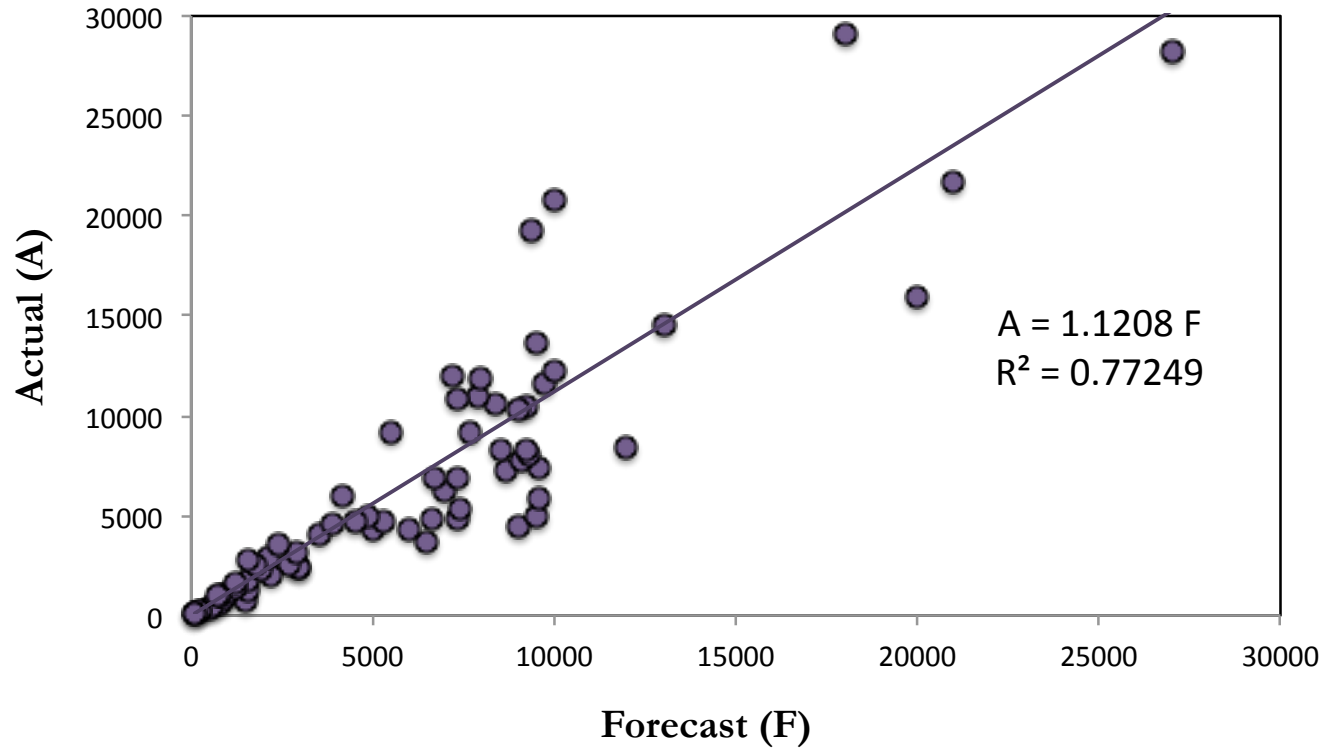
$$\text{A/F Ratio} = \frac{\text{Actual Demand}}{\text{Forecast}}.$$

“The frequency distribution of past forecast errors was then used as a probability distribution for the as yet unrealized future forecast errors.”

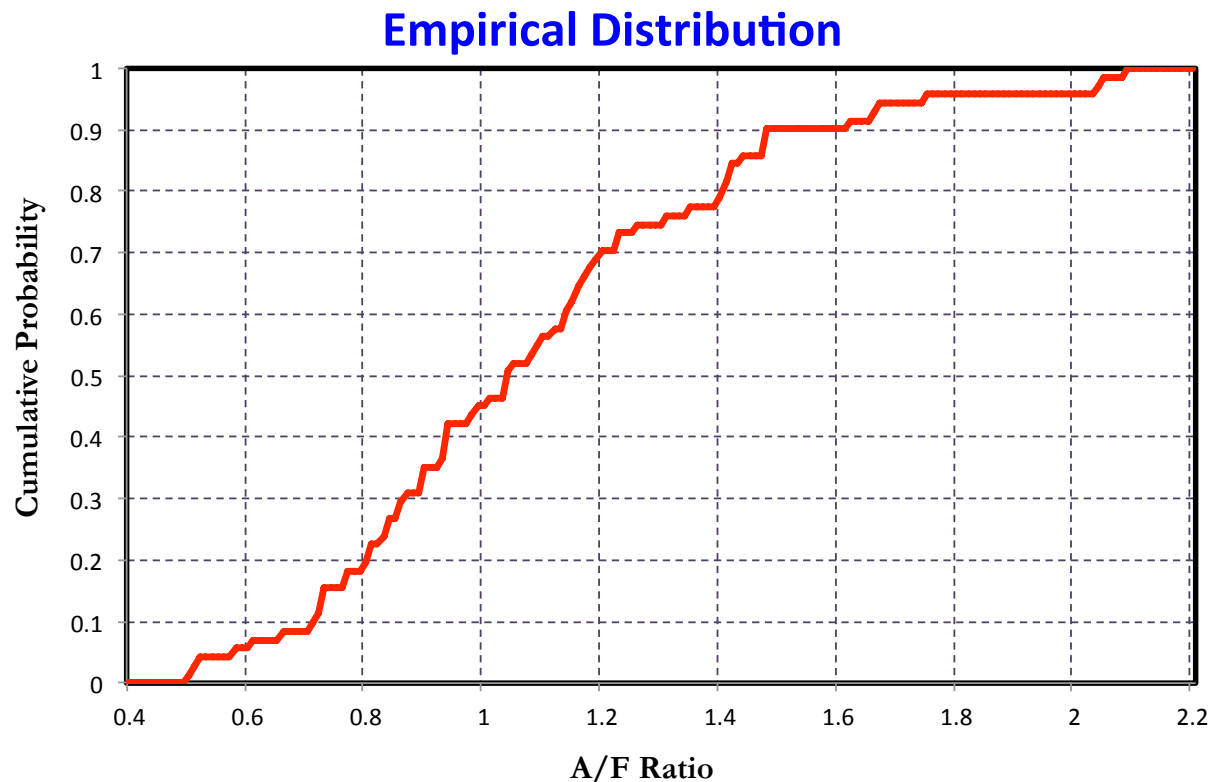
– quote from L.L. Bean case.

ITEM	FORECAST (a)	ACTUAL (b)	A/F (a/b)	Rank (c)	F(A/F) (c / 71)
Wool/Cashmere Blazer	9000	4448	0.49	1	0.0141
Heat Factory Gloves	1500	754	0.50	2	0.0282
⋮	⋮	⋮	⋮	⋮	⋮
Travel Umbrella	1200	1394	1.16	47	0.6620
Sunwashed Canvas Shorts	3900	4602	1.18	48	0.6761
⋮	⋮	⋮	⋮	⋮	⋮
Floral Print Swim Dress	9400	19216	2.04	70	0.9859
Seersucker Knit Shirt	10000	20818	2.08	71	1.0000

L.L. Bean Forecast versus Actual



L.L. Bean Solution: Using Empirical Distribution



Parameters:

- ✧ Price \$45
- ✧ Wholesale cost \$25
- ✧ Discounted sales price \$15
- ✧ Forecast 12,000 units

Issues Not Captured by the Newsvendor Model in the L.L. Bean Case

- **Data:** Ensuring that demand information is captured even when they run out of stock.
- **Liquidation Price:** this will most likely depend on the amount of inventory left over
- **Intangible Costs:**
 - ✓ Effect of a lost sale on future sales to a customer
 - ✓ Possibility of stealing a competitor's customer by having additional stock of a product.
- **Substitution Effects:** some customers are likely to buy a different item (color or product) when one item is stocked out.
 - ✓ If we realize there is substantial substitution, how should we change our inventory decisions?

L.L. Bean Key Lessons

- ✓ The mismatch between supply and demand is costly: \$21M on sales \$528M (4%)
- ✓ Systematic methods for choosing order quantities are needed when managing hundreds or thousands of items.
- ✓ Implementation of the Newsvendor model using data on historical forecast errors, i.e. you must quantify your uncertainty.
- ✓ The profit maximizing order quantity generally does not equal expected demand.
- ✓ Separate the forecasting process from the ordering process.

Module 6: Key Ideas

➤ Newsvendor:

- ✓ Demand is uncertain (random) and there is a single ordering opportunity.
- ✓ There is a “too much-too little” challenge.
- ✓ Optimal solution balances the cost and likelihood of stockouts and excess inventory.

✓ **Critical Fractile solution:**

$$\text{Prob}(D \leq Q^*) = \frac{C_s}{C_s + C_e}$$

➤ Economic Order Quantity.

- ✓ Demand and inventory replenishment are fairly stable and predictable.
- ✓ Optimal solution balances inventory and ordering costs.

✓ **EOQ solution:**

$$Q^* = \sqrt{\frac{2 \lambda s}{h}}$$

➤ The (Q,R) model:.

- ✓ Combines the EOQ and Newsvendor Solution

✓ **(Q,R) solution:**

$$Q^* = \sqrt{\frac{2 \lambda s}{h}} \quad \text{and} \quad R^* = \lambda L + Z_{SL} \sigma \sqrt{L}$$

After Class Activities

➤ REVIEW SESSION 3

➤ GROUP:
WORK ON REPORTS AND PRESENTATION

Beer Game Simulation: Bring your laptop to class